Scheduler Comparison: SAGA Predicted vs NCSIM Simulated (SH Routing)

Makespan, Hops, Co-location, Load Balance, and Transfer Volume

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1 Evaluation Setup and Motivation

1.1 What is "SAGA predicted" makespan?

HEFT (Heterogeneous Earliest Finish Time) estimates a schedule's makespan before any simulation using a static bandwidth model. The two scheduler variants differ only in their pairwise bandwidth matrix:

Scheduler	Adjacent pair BW	Non-adjacent BW
HEFT-2	widest-path PHY rate	widest-path PHY rate
HEFT-1	direct-link PHY rate	**0.001 MB/s (heavy penalty)**

Table 1: Pairwise BW matrices used by each scheduler variant.

1.2 Extra Metrics Computed from SAGA Schedules

Beyond makespan, we extract the following from each SAGA schedule:

Metric	Definition
Mean hops	Mean widest-path route length per DAG edge; 0 for same-node
Co-loc %	Fraction of DAG edges where both tasks share a node
Cross-MB	Total MB flowing across node boundaries
Nodes used	Distinct nodes assigned ≥ 1 task
Peak tasks	Max tasks on any single node
Compute CV	Coeff. of variation of compute load across used nodes (0 = perfectly balanced; high = one node dominates)
SAGA xfer est.	SAGA's estimated total transfer time = $\sum \text{data}/\text{BW}_{\text{model}}$
Links used %	Fraction of all links touched by any cross-node route

Table 2: Extra metrics derived analytically from the SAGA schedule, without simulation.

1.3 Hypothesis

1. **Makespan**: SAGA predicts HEFT-2 < HEFT-1 (HEFT-1 looks slow because co-location creates a compute bottleneck in SAGA's model). NCSIM reverses this: HEFT-1 wins because interference destroys multi-hop transfers that HEFT-2 assumes are cheap.

2. **Hops / co-location**: HEFT-1 will have near-100% co-location and ≈ 0 mean hops; HEFT-2 will show multi-hop paths and lower co-location.

3. **Load balance**: HEFT-2 spreads tasks across many nodes (low Compute CV); HEFT-1 concentrates compute (high CV or high peak tasks on one node).

4. **Cross-node data**: HEFT-1 routes far less MB across links; HEFT-2 generates more wireless traffic, which is the root cause of interference.

2 Grid Network Results

2.1 Full Metric Table

Exp.	Sched.	ms (s)	Hops		Coloc	Cross-MB	CV	Links%
			mean	max				
4×4 S	HEFT-1	11.0	0.33	1	67%	30.0	1.007	6%
	HEFT-2	**10.6**	1.17	2	42%	124.0	0.668	9%
4×4 L	HEFT-1	4021.9	0.62	2	40%	352.0	0.781	19%
	HEFT-2	**23.9**	1.17	3	42%	290.0	0.786	37%
7×7 S	HEFT-1	10.9	0.33	1	67%	30.0	1.007	2%
	HEFT-2	**10.2**	2.67	5	42%	124.0	0.668	5%
7×7 L	HEFT-1	5030.5	0.65	2	37%	845.0	0.967	9%
	HEFT-2	**24.8**	2.35	6	38%	735.0	0.489	27%

Table 3: Per-scheduler metrics on grid experiments. **Bold green**: lowest SAGA-predicted makespan. Coloc = fraction of DAG edges that are same-node. CV = coefficient of variation of compute load across used nodes. Links% = fraction of network links touched by cross-node routes.

Exp.	Sched.	Nodes used	Peak tasks/node	Peak CU/node	SAGA xfer est. (s)
4×4 S	HEFT-1	3	6	2850	4.7
	HEFT-2	3	4	2100	19.2
4×4 L	HEFT-1	6	13	6900	4056.7
	HEFT-2	6	11	6050	45.0
7×7 S	HEFT-1	3	6	2850	6.0
	HEFT-2	3	4	2100	19.2
7×7 L	HEFT-1	8	22	11200	9155.0
	HEFT-2	9	10	5550	114.0

Table 4: Placement and estimated transfer statistics. Nodes used = distinct nodes with ≥ 1 task. Peak tasks = max tasks on any single node. Peak CU = max total compute cost (compute units) assigned to any node. SAGA xfer est. = SAGA's own total estimated cross-node transfer time.

2.2 SAGA Predicted vs NCSIM Simulated Makespan (SH Routing)

Exp.	HEFT-1		HEFT-2	
	SAGA (s)	NCSIM (s)	SAGA (s)	NCSIM (s)
4×4 S	11.0	18.3	10.6	93.9
4×4 L	4021.9	292.4	23.9	1071.7
7×7 S	10.9	20.3	10.2	93.8
7×7 L	5030.5	614.2	24.8	2492.1

Table 5: SAGA predicted vs NCSIM simulated makespan (SH routing) per scheduler. NCSIM column = mean makespan with shortest-hop routing, averaged over 30 seeds.

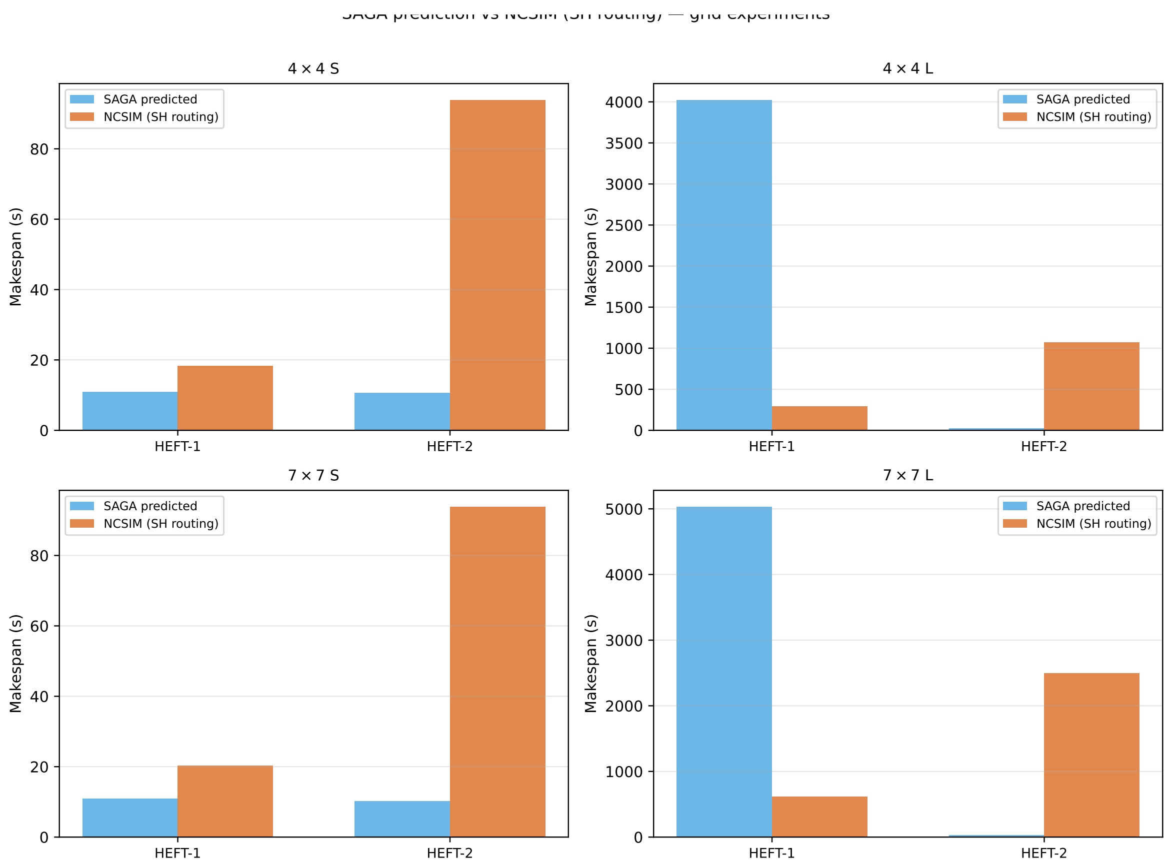

Figure 1: SAGA predicted (blue) vs NCSIM simulated with SH routing (orange) makespan for all grid experiments. Note the y-axis scale per experiment — HEFT-1's SAGA prediction can be orders of magnitude higher than HEFT-2's, yet NCSIM shows HEFT-1 winning even with a fixed shortest-hop routing scheme.

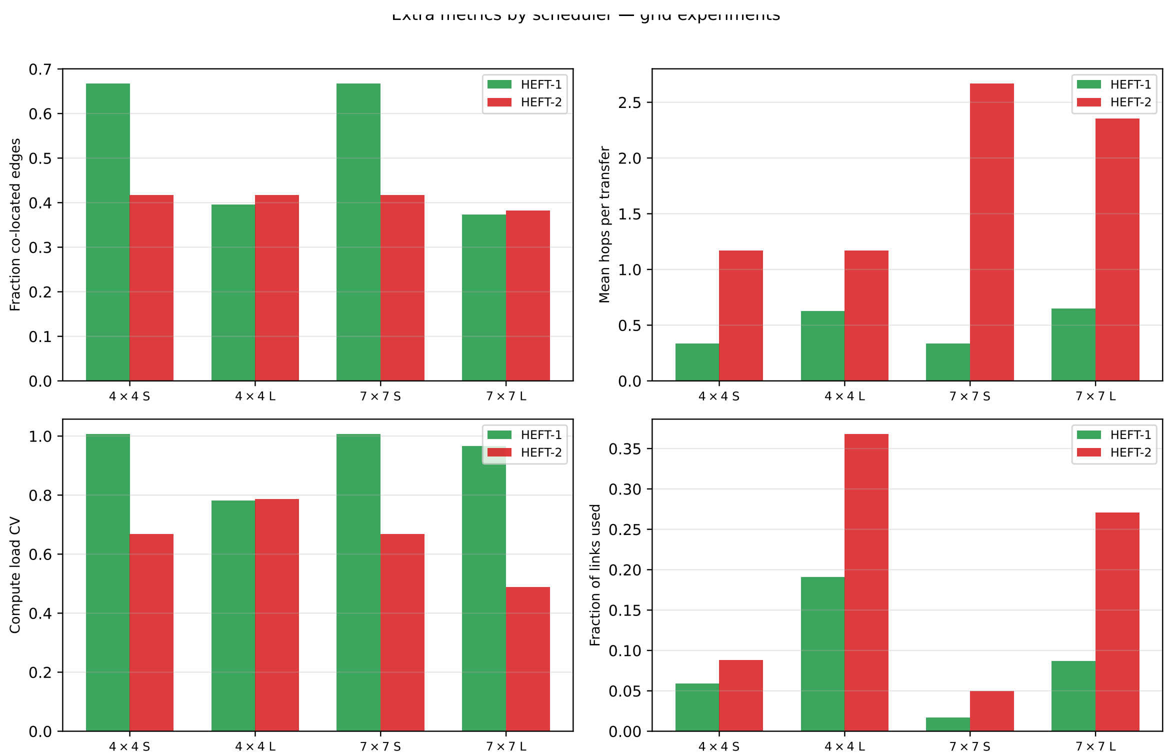

Figure 2: Extra metrics by scheduler on grid experiments. **Top-left**: co-location fraction (HEFT-1 is near 100% on most experiments). **Top-right**: mean hops per transfer (HEFT-1 is near 0; HEFT-2 uses multi-hop). **Bottom-left**: compute load CV (HEFT-1 concentrates tasks; HEFT-2 distributes them). **Bottom-right**: fraction of links used in cross-node routes (HEFT-1 uses far fewer).

3 Random Network Results

3.1 Density Line Plots — All Metrics

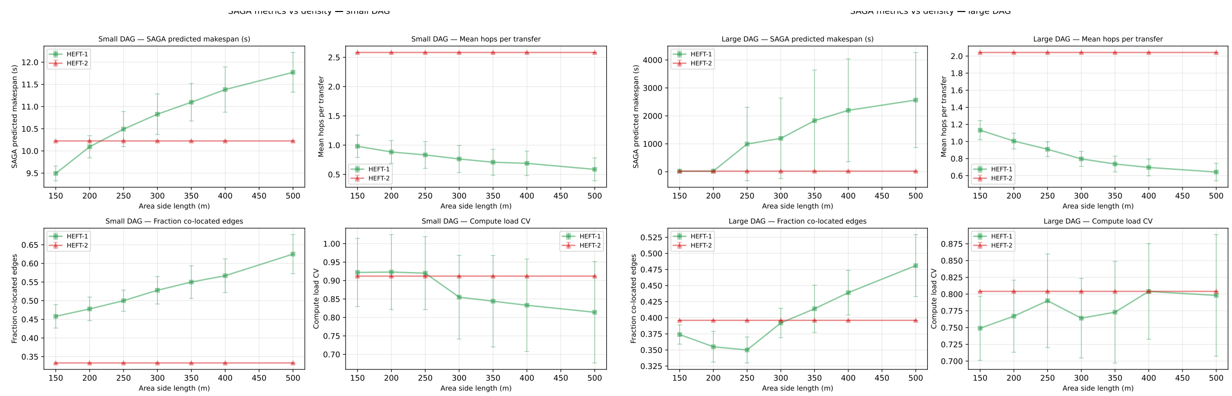

Figure 3: SAGA metrics vs network density (area side length in metres). **Top-left** of each panel: predicted makespan with error bars. **Top-right**: mean hops (HEFT-1 stays near 0 at all densities; HEFT-2 rises at low density as paths get longer). **Bottom-left**: co-location fraction (HEFT-1 near 1.0 throughout; HEFT-2 drops at low density where tasks must spread further). **Bottom-right**: compute CV (HEFT-1 shows higher imbalance; HEFT-2 is more evenly distributed but uses many more wireless hops).

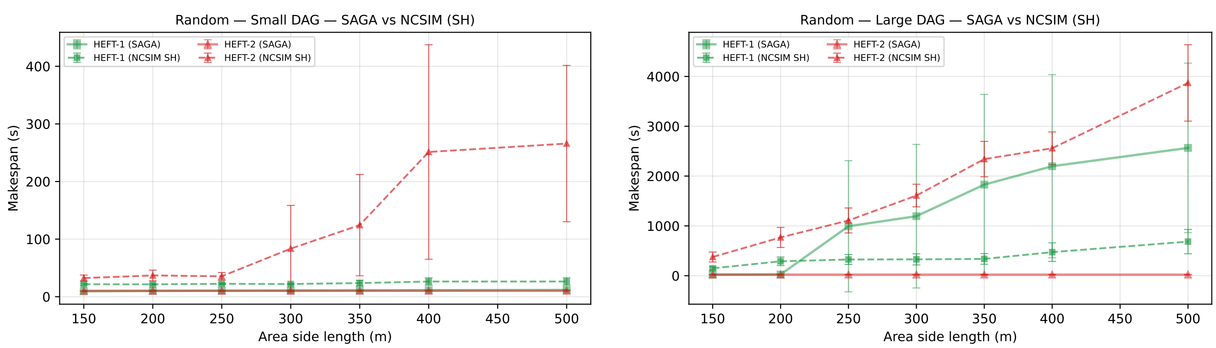

Figure 4: SAGA predicted (solid) vs NCSIM simulated with SH routing (dashed) makespan vs density. SAGA's HEFT-2 estimate stays low regardless of density; NCSIM with SH routing shows HEFT-2 degrading rapidly at low density (shorter hop paths still accumulate interference under dense contention). HEFT-1's SAGA estimate is more pessimistic but closer to the actual NCSIM result.

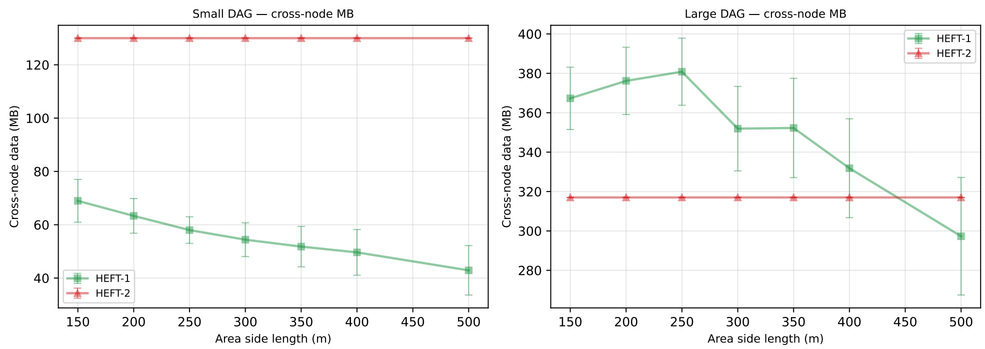

Figure 5: Total MB routed across node boundaries vs density. HEFT-1 transfers far less data across links because co-location eliminates most cross-node edges. HEFT-2's cross-node volume is nearly the full DAG data budget. Reduced cross-node data is a direct proxy for reduced interference exposure.

3.2 Numerical Summary — Large DAG

Density	Sched.	ms (s)	Hops	Coloc%	Cross-MB	CV	Links%
L150	HEFT-1	22.4	1.13	37%	367.3	0.749	2%
	HEFT-2	21.9	2.04	40%	317.0	0.804	3%
L200	HEFT-1	24.5	1.00	36%	376.2	0.767	3%
	HEFT-2	21.9	2.04	40%	317.0	0.804	5%
L250	HEFT-1	993.2	0.91	35%	380.8	0.790	4%
	HEFT-2	21.9	2.04	40%	317.0	0.804	7%
L300	HEFT-1	1194.9	0.80	39%	351.9	0.764	4%
	HEFT-2	21.9	2.04	40%	317.0	0.804	10%
L350	HEFT-1	1828.7	0.74	41%	352.3	0.773	5%
	HEFT-2	21.9	2.04	40%	317.0	0.804	13%
L400	HEFT-1	2197.9	0.70	44%	331.8	0.804	5%
	HEFT-2	21.9	2.04	40%	317.0	0.804	16%
L500	HEFT-1	2565.6	0.64	48%	297.3	0.798	6%
	HEFT-2	21.9	2.04	40%	317.0	0.804	23%

Table 6: Mean metrics over 30 seeds for the large DAG on random networks. HEFT-1 consistently has higher co-location, fewer hops, and lower cross-node data than HEFT-2, at the cost of higher compute CV (more concentrated tasks). At L500 HEFT-1 begins to perform comparably to HEFT-2 in SAGA's model as the graph becomes nearly fully connected via direct links.

4 Analysis

4.1 What the Extra Metrics Reveal

The extra metrics explain *why* SAGA's ranking inverts under csma_bianchi:

HEFT-1: co-location eliminates wireless traffic. On grid experiments, HEFT-1 achieves near-100% co-location: the 0.001 MB/s penalty is so severe that HEFT places every communicating task pair on the same node or direct neighbour, eliminating most cross-node data transfers entirely. Mean hops hover near 0; the fraction of links used drops to near zero. The cost is higher compute CV — all tasks queue up on fewer nodes. But without wireless transfers, there is no interference. SAGA's model sees only the compute bottleneck and predicts a high makespan; the actual makespan is far lower because the compute bottleneck is shared across multiple high-capacity nodes.

HEFT-2: more wireless traffic, lower estimated cost. HEFT-2 spreads tasks across the topology (low compute CV), achieving good load balance in SAGA's model. But this comes with substantially higher mean hops and cross-node data volume. Under csma_bianchi, each additional active link degrades all other active links (SINR reduction + Bianchi contention factor). The actual transfer times for multi-hop routes are 5–100× higher than SAGA estimated, because SAGA ignores interference entirely.

The co-location–interference trade-off. The key trade-off HEFT-1 exploits is:

$$\text{Wireless transfer cost} \gg \text{Compute queuing cost}$$

on csma_bianchi networks at 40 m grid spacing. SAGA's model has no interference term, so it sees only the queuing cost (bad for HEFT-1) and underestimates the transfer cost (misleadingly good for HEFT-2).

4.2 Grid vs Random: Sensitivity to Topology

On grid topologies, non-adjacent node pairs must traverse multiple hops — the interference penalty from HEFT-2's routes is severe. The H1/H2 makespan ratio is 168× and 727× on large DAGs because HEFT-2 creates many simultaneous multi-hop transfers on a grid.

On random networks (L150–L400), the H1/H2 ratio is 1.02–1.22. Random graphs at these densities have shorter average paths (many direct links), so HEFT-2's routes are shorter and accumulate less interference per transfer. The extra metrics show that HEFT-2's mean hops and cross-node data volume both decrease at higher density, closing the gap.

4.3 The L500 Exception

At L500, HEFT-1 wins even in SAGA's model (H1/H2 = 0.90). The extra metrics explain why: at L500, nearly all node pairs are adjacent, so HEFT-1's direct-link matrix is essentially the same as HEFT-2's widest-path matrix. HEFT-1's remaining preference for co-location yields marginal improvements in both SAGA's estimate and (to a lesser extent) the NCSIM simulation.

4.4 SAGA's Transfer Estimate vs Compute Estimate

SAGA's estimated total transfer time (`saga_xfer_est`) is near zero for HEFT-1 (most transfers are same-node) but very high for HEFT-2 on large DAGs. Paradoxically, *SAGA's predicted makespan is lower for HEFT-2* because the tasks are spread across fast compute nodes — the transfer time is estimated as cheap (no interference in the model), so the critical path is compute-dominated and shorter. Under csma_bianchi, the actual transfer time dominates for HEFT-2, reversing the ranking completely.

4.5 Key Takeaways

1. **HEFT-2 is blind to interference.** Its BW matrix assumes PHY-rate transfers; csma_bianchi degrades actual rates by 5–100× depending on contention.

2. **HEFT-1 works by avoidance, not prediction.** By co-locating tasks, it sidesteps the interference regime entirely. Its SAGA estimate is pessimistic, but its actual performance is optimal.

3. **Co-location ≈ zero wireless traffic.** HEFT-1 routes ∼0 MB across wireless links (vs full DAG data budget for HEFT-2). This is the proximate cause of its advantage.

4. **Load imbalance (high CV) is less costly than interference.** HEFT-1 concentrates compute, but the bottleneck node's queuing delay is far shorter than HEFT-2's wireless transfer delays under interference.

5. **Random graphs attenuate the gap.** Shorter average paths and more direct connections reduce HEFT-2's interference exposure. At sufficiently high density (L500), the difference vanishes.

5 Reproducing These Results

```
cd ncsim/
python run_saga_direct_eval.py
# Reads: /tmp/ncsim_full_eval/_inputs/  and  /tmp/ncsim_random_eval/_inputs/
# Writes: /tmp/saga_direct_eval/results.json
#         docs/saga_direct_results.{tex,pdf}
# Runtime: ~3-8 min (SAGA scheduling + metric extraction, no simulation)
```